

pq402

An API that charges an AI agent, checks it is allowed to buy,
and never learns which buyer it was

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<https://github.com/Galmanus/pq402>

Abstract

Autonomous software is starting to buy things: data, API calls, compute. The protocols for that are arriving — x402 revives HTTP’s long-dormant **402 Payment Required** and settles it on-chain — but they answer only one question, *did the agent pay*. Metered services have a second, *is this agent allowed to buy this at all*, and today’s answers to it are an API key or an account, both of which stamp the buyer’s identity onto every request they will ever make.

This paper describes a working system that answers both, and answers the second without learning who is asking. The eligibility check is a hash-based zero-knowledge proof judged by a Soroban contract, not by the server; a per-challenge nullifier is burned in contract storage, so single use is enforced by consensus and survives the server being restarted, replaced, or dishonest. In its unlinkable mode the proof publishes a commitment rather than a credential identifier, with fresh blinding on every use: the server learns that *a member of the issuer’s set* paid and nothing further, and two payments by one credential share no public value, so they cannot be linked to each other.

Everything reported here runs on Stellar testnet and every claim is backed by a transaction hash. We also record four undocumented details of the x402 protocol on Stellar that each cost a debugging round to find, and publish the fix as a package so the next team does not pay for them again. The limits are stated with the same precision as the results: the proof system is post-quantum, the transaction carrying it is signed with Ed25519 and is not.

1. The gap

An agent that buys something must be authorised to buy it. The two mechanisms in use today both work by identifying the buyer.

An **API key** is a bearer identifier. Every call carries it, so the seller accumulates a complete, joinable history of one buyer’s behaviour — which is usually not what either party wanted, merely what was easy to build.

An **account** is worse for the buyer and better for the seller, and has the same shape: eligibility is established by being known.

Neither is a failure of care. There has not been an alternative that a server could check cheaply, that a client could produce quickly, and that did not require the server to be trusted with the answer. What follows is an attempt at one.

1.1 Why now

Three things had to be true at once, and only recently are:

- **A payment rail an agent can drive.** x402 defines a 402 response carrying machine-readable payment requirements, and on Stellar the client signs Soroban *authorization entries* rather than transaction envelopes — so a facilitator assembles the transaction and pays the network fee. An agent needs the asset it intends to spend and no XLM at all.
- **A proof cheap enough to verify inside a smart contract.** Circle STARKs over the Mersenne-31 field are the cheapest hash-based proof system in production. Verification fits an ordinary Soroban contract call.
- **Somewhere to put the single-use guarantee that is not the server.** Contract storage. A nullifier burned there cannot be forgotten by restarting a process.

1.2 Where this sits in Stellar’s own quantum plan

On 4 August 2026, days before this was written, the Stellar Development Foundation published its Quantum Preparedness Plan.¹ It is a signature-migration plan first: ML-DSA-44 and ML-DSA-65 as Soroban host functions in 2026, quantum-safe signers on classic accounts in 2027, Ed25519 retired when the threat demands it. None of that is what this work does; account authentication is a different layer, and the disanalogy should be stated rather than blurred.

The plan is candid about what it does not cover, and that paragraph is this work’s exact address:

“The QPP does not yet address pairing-based zero-knowledge protocols built on Stellar. . . there is no drop-in post-quantum replacement for pairing-based SNARKs with comparable performance. STARKs and lattice-based proof systems are candidates, each with significant trade-offs. As part of QPP we will convene ZK protocol teams. . . to develop a shared research agenda.”

Two of the plan’s own words carry the point. The zero-knowledge layer is the *unfinished* half of Stellar’s quantum migration, and STARKs are one of the two *named* candidates for it. This system is a working instance of that candidate — a Circle STARK verified inside a Soroban contract — and the “significant trade-offs” the plan gestures at are the ones the Measurements section puts numbers on: a 196 KB proof filling 86% of the transaction envelope, roughly 96 conjectured bits of soundness, and about twice the instruction cost of a pairing. That is not an endorsement of this work by the foundation, which the plan does not give and this paper does not claim. It is a data point of exactly the kind the plan says is missing, arriving as the foundation asks for them.

2. What the system does

A request to a paid route meets two independent gates. Neither is decided by the server, and the server cannot overrule either.

gate	mechanism	who decides
payment	x402 exact , USDC over its Soroban Asset Contract	Stellar consensus
credential	hash-based STARK, nullifier burned in storage	Stellar consensus

The flow, in the order it happens:

¹stellar.org/blog/foundation-news/introducing-the-quantum-preparedness-plan

```

agent -> GET /premium
      <- 402 payment-required header + credential challenge

agent proves eligibility locally, 7 ms
agent -> POST /pq/unlock { proof, publics }
server-> Soroban: verify the proof and burn the nullifier, one transaction
      <- pass, and the transaction hash of the burn

agent -> GET /premium + PAYMENT-SIGNATURE + the pass
server-> facilitator: verify, then settle
      <- 200, the goods, and the settlement hash

```

Two orderings in that sequence are deliberate and worth naming.

The credential is checked before the payment. Charging someone and then refusing them is worse than refusing them, and arranging that is the server’s fault, not the protocol’s.

Verification precedes settlement. `settle` moves money, so a payload that fails verification must never reach it. The package has a test asserting that a rejected verification leaves `settle` uncalled, because the ordering is the only thing standing between a bad payload and a real transfer.

3. The credential, and what the server learns

3.1 Identifying mode

The relation proves, in zero knowledge, that the caller knows a secret s such that

$$\text{leaf} = \text{trunc}_8(\pi(s \parallel \text{action})), \quad \text{nullifier} = \text{trunc}_8(\pi(s \parallel \text{round}))$$

share that one secret, where π is the Poseidon2 permutation over Mersenne-31. The AIR constrains the secret cells of the two permutation blocks to be equal in the same row; without that constraint a prover could present one member’s leaf beside another member’s nullifier.

Both published values are *truncated* to a digest. An earlier version published the full sixteen-limb permutation state, and that was a break rather than a wart: π is a bijection and the context sits public beside it, so a single inversion returned the secret. It was found by audit, demonstrated against a real testnet proof in milliseconds, and fixed by publishing digests — which puts recovery back at $|F|^8 \approx 2^{248}$.

In this mode the leaf is a public input. The server therefore knows *which* credential paid, and two payments by one credential are trivially linked. That is anonymous in the sense of not knowing your name; it is not anonymous in the sense of not knowing which of the thousand you are.

3.2 Unlinkable mode

The second mode closes that. The relation publishes

$$C = \text{compress}(\text{leaf} \parallel \text{blinder})$$

with a fresh blinder drawn from system entropy on every use, and never the leaf. Two proofs compose on that shared C :

- the **acting** half proves that a valid credential acted under C , and burns its nullifier;
- the **membership** half proves that C sits under the issuer’s published Merkle root, through a private authentication path.

What the server learns is exactly: *someone in the issuer's set paid*. Not which member. And because the blinder is fresh, two payments by one credential share no public value at all, so they cannot be linked to each other either.

Three orderings again, each for a reason:

- the two commitments are compared **before** anything is spent — two proofs are only about one credential if they share C , and checking that is free while the proofs cost two transactions;
- membership is verified **before** the acting half — membership is a read-only call and acting burns a nullifier, so there is no reason to burn one for a caller who was never in the set;
- the claimed root is compared against the **issuer's** — a membership proof is only ever a statement about the tree the prover chose, and without that check anyone can build a tree containing their own leaf.

3.3 The challenge is not the server's to choose

The acting contract derives the expected round from the ledger sequence:

$$\text{epoch} = \left\lfloor \frac{\text{ledger_sequence}}{720} \right\rfloor$$

and refuses any other. Freshness is therefore guaranteed by consensus rather than asserted by whoever is asking, and one credential acts once per epoch — about an hour at Stellar's close time. A server cannot replay an old challenge at a member, and a member cannot pick a convenient one.

4. Four things about x402 on Stellar that are not written down

Each of these cost a debugging round, and each produced an error message that pointed somewhere other than the cause. They are encoded as behaviour in the published package rather than as advice, because a comment does not stop anyone from getting it wrong.

1. **v2 carries the requirements in a payment-required header, not the body.** The body is read only when `x402Version == 1`. A server that publishes perfect requirements as JSON alone is invisible to a v2 client, which fails with *Invalid payment required response* having never looked at them. Every 402 a gated route emits needs the header — including the one meaning “you have a session but still owe payment”, which is the one everybody forgets.
2. **The signed payload returns in PAYMENT-SIGNATURE**, with X-PAYMENT the v1 name. Read both.
3. **extra.areFeesSponsored must appear in the requirements** or the Stellar exact scheme refuses to build a payment at all. That flag is the facilitator's assertion, so it should be fetched from `/supported` at startup rather than claimed on the facilitator's behalf.
4. **Stellar assets carry seven decimals, not six.** `0.10` is 1000000 base units. The six an EVM habit expects is a tenth of the intended price, settling quietly — the worst kind of bug, because it works.

A fifth, adjacent: the Stellar CLI's `-network` takes a configuration *name* (`testnet`) while x402 takes a CAIP-2 *identifier* (`stellar:testnet`). Feeding one to the other fails with *Invalid name*, four frames from anything that mentions networks.

5. Spending limits that are rules, not promises

A client-side cap — `--max 0.10` on the agent — is a promise the agent makes to itself. An agent that has been prompt-injected, misconfigured, or handed a bad tool description simply stops making it, and nothing outside the agent notices.

`agent-treasury` is the same limit as a rule. It is a Soroban contract holding the agent’s funds, enforcing two policies before any token moves:

- a **contract allow-list**, so a stolen policy key cannot be pointed at a token the treasury was never meant to touch;
- a **rolling daily cap**, keyed off ledger time rather than a counter that calling more often can reset.

Both refusals happen in consensus. On testnet: a payment inside policy settles; one over the cap is refused with error `#3`; one naming a token outside the allow-list is refused with `#2`; and the remaining budget is unchanged after both, because a refusal consumes nothing.

The budget is committed *before* the transfer: if the transfer panics the whole invocation reverts, so the ordering cannot leak budget, while committing afterwards would leave a window in which a reentrant call still sees the old total.

6. Measurements

All on Stellar testnet, and each number was measured rather than estimated.

proving, identifying mode (median of 15 runs, process start included)	7 ms
proving, range across those runs	6–11 ms
proof size, identifying mode	79,186 B
proof size, unlinkable mode (acting half)	113,360 B
proof size, unlinkable mode (membership half)	82,765 B
on-chain verification, unlinkable act (12 queries, blowup 2 ⁷)	243,213,882 instr.
on-chain verification, identifying spend (40 queries, blowup 2)	265,797,092 instr.
Soroban per-transaction instruction ceiling	400,000,000 instr.
transaction envelope, unlinkable act	114,084 B
transaction envelope, identifying spend	77,508 B
Soroban per-transaction size ceiling	132,096 B
settlement fee, paid by the facilitator	23,073 stroops

The instruction rows are read from the submitted envelopes’ declared Soroban resources, which the CLI sets from simulation; the transactions are the ones linked in the repository and both succeeded.

Two things fall out of them. The first is which ceiling actually binds, and it is not the one we expected. Instructions sit at 61% of budget, which sounds like room; the envelope sits at **86%** of the 132,096-byte transaction size cap, which is not room at all. The proof has to travel to the chain before it can be checked there, and on Soroban the trip is scarcer than the check. Every reference to a “400M instruction cap” as the limiting resource, including two we made earlier in this work, was measuring the wrong wall.

The second is worth pausing on. The identifying gate asks 40 queries at blowup 2 and costs *more* — 265.8M against 243.2M — while being the weaker of the two by a wide margin. Blowup buys soundness at a discount that queries do not: raising the rate lengthens the committed codeword, which the prover pays for, while the verifier’s cost tracks the number of openings it has to check. On a chain that meters the verifier and not the prover, few queries over a high blowup wins on both cost and soundness.

It does not win on everything, and the obvious next move is a trap. Pushing the blowup higher again would buy more soundness for a few percent more instructions, and would quietly destroy the hiding property. Counting the FRI phase as a linear-algebra problem — what the verifier observes against what the blinding can cover — gives observations $Q \cdot \ell$ with $\ell = \log_2(2N \cdot 2^\beta)$, against $3N$ dimensions of blinding: the trace blinder, collapsed to N by the α -reduction, plus the $2N$ of the randomisation polynomial. So soundness improves as $Q\beta$ while the zero-knowledge margin degrades as $Q(\log_2 2N + \beta)$, and zero-knowledge loses the race because it pays the committed height on top of the blowup. A negative margin is conclusive in the bad direction: fewer blinding dimensions than revealed scalars means some linear functional of the witness survives into the transcript, and no simulator of this shape exists.

This is why the contracts *pin* the configuration rather than enforce a floor. A floor was the first fix, and it was the wrong one: a caller passing a larger blowup would have been accepted as stronger while silently destroying hiding. There is no scalar to be above — there is one configuration checked on all four axes at once, and anything else is a different configuration that has not been. The lever that helps both directions is trace height, which adds three blinding dimensions per row against Q observations per doubling.

And that is where the four constraints close on each other, which is the result this section exists to report. The deployed configuration — 128 rows, 12 queries, blowup 2^7 — sits at 92 conjectured bits of soundness, a FRI zero-knowledge margin of +204, 61% of the instruction budget, and 86% of the envelope. Each available move breaks one of the other three. Raising the blowup buys soundness and drives the margin toward zero, which is conclusive against hiding. Raising the queries buys soundness and adds a Merkle path per query to an envelope already at 86%. Raising the trace height is the only move that helps soundness and hiding together, and it is precisely the move that doubles the proof and overruns the size cap first. The configuration is not a preference. It is the last point that satisfies all four at once, and the reason it looks arbitrary is that three of the four walls are invisible until measured.

For scale, SDF’s own privacy-pool prototype spends roughly 40M instructions per BLS12-381 pairing. A Groth16 verification needs several, so a hash-based verifier here costs on the order of twice a pairing-based one. That factor of two is the whole price of dropping the trusted setup and the exposure to Shor.

The fee row is the one worth pausing on: the account charged the network fee is the facilitator’s, not the agent’s. That is fee sponsorship working, and it is why an agent wallet needs only the asset it intends to spend.

6.1 Security parameters

FRI soundness is approximately queries $\times \log_2$ (blowup) bits before the proof-of-work grind, and it cannot exceed the field the challenges are drawn from. Both halves of that sentence matter.

The identifying gate originally ran at 40 queries with blowup 2: about 48 bits conjectured, and near 24 against a Grover-assisted adversary, since Fiat-Shamir grinding falls to a square-root search. Twenty-four bits is a demonstration parameter wearing a production label.

Raising it on that path was impossible rather than merely expensive: its challenge field is a degree-3 extension of Mersenne-31, about 93 bits, so 46 quantum bits is that path’s ceiling at any configuration. The gate now used draws challenges from QM31, a degree-4 extension with a 124-bit ceiling, and runs at 16 queries with blowup 2^6 : roughly 96 conjectured bits, near 48 quantum. The configuration is pinned inside the contract rather than passed by the caller, because whoever picks the security level picks how hard their own proof is to forge.

7. Reusability

The protocol work is published, not merely described.

```
npm install x402-stellar-paywall
```

```
const pay = await expressPaywall({ price: "0.10", payTo: RECIPIENT });  
app.get("/weather", pay, (req, res) =>  
  res.json({ temp: 24, settled_by: req.x402.transaction }));
```

That is the whole integration. Adding `credential: { ... }` adds the eligibility gate; adding `mode: "crowd"` to it makes that gate unlinkable. Bindings exist for Express, Hono and FastAPI — the three frameworks the sub-lane names — over a framework-free core with no runtime dependencies.

It was proven somewhere other than its birthplace: `examples/second-app` is a weather API with no credential logic in it at all, which declares the package as an ordinary dependency, gates one route with one line, and was paid by an agent on testnet.

8. Related work

None of the cryptography here is new, and it would be dishonest to present it as such. Post-quantum anonymous credentials are a studied object with a literature, and this system is an engineering result standing on it. The comparison worth making is not against nothing; it is against the published numbers.

Policharla, Westerbaan, Faz-Hernández and Wood² build post-quantum Privacy Pass from post-quantum anonymous credentials, proving possession of a zkDilithium signature — a STARK-friendly variant of Dilithium2 — inside a STARK. Their tokens run 85–175 KB at a 115-bit proof security level, generate in 0.3–5 s, and verify in 20–30 ms. That is the closest published relative of what runs here, and it is the reference a reader should hold this against.

Chen et al.³ give the first post-quantum Direct Anonymous Attestation from symmetric primitives alone, a SPHINCS+-derived credential shown through an MPC-in-the-Head proof. Feneuil and Rivain’s CAPSS⁴ builds signatures from arithmetization-oriented permutations — Rescue-Prime, Poseidon, Griffin, Anemoi — and reports anonymous-credential presentation proofs under 150 KB at 128 bits. That is the same primitive family this work uses, and the better-optimised end of it. On the lattice side, Argo et al.⁵ reach credentials of 80 KB, since improved to 54 KB.

Set against those, our proofs are 79 KB in identifying mode and 196 KB across both halves of the unlinkable one, at roughly 96 conjectured bits. That is larger and weaker than CAPSS, and it is worth saying plainly. It is also precisely where the theory said it would land: Jeudy et al.⁶ note in passing that anonymous credentials from STARKs are possible but that the cost “appears to be in the hundreds of kilobytes range.” We landed at 196 KB, which is the footnote coming true rather than a surprise.

What is new is not the credential. It is where the verifier runs. Every scheme above is verified in software, by the party being shown the credential — a Privacy Pass origin, a TPM verifier, a relying party. Verification is a claim about that party’s honesty. Here the verifier is a Soroban contract, so the verdict is a fact about the ledger: the nullifier burns in consensus storage, and the round the proof answers is derived from the ledger sequence rather than chosen by whoever is asking. Nobody has to trust the server, because the server is not the one deciding.

That relocation is also what explains the gap in the numbers. Soroban’s per-transaction limits, not the cryptography, are what hold the security parameters down — and the binding one is the 132,096-byte envelope rather than the instruction budget, as the measurements below

²Cryptology ePrint 2023/414, ia.cr/2023/414.

³ePrint 2024/650, PQCrypto 2023.

⁴ePrint 2025/061, with the SmallWood proof system at ePrint 2025/1085.

⁵ePrint 2024/131, ACM CCS 2024.

⁶ePrint 2023/560.

show. A software verifier with 20–30 ms to spend has no such ceiling and no need to ship the proof anywhere. Closing that gap means smaller proofs rather than a faster verifier, which is exactly what CAPSS and SmallWood are for, and is where we would start.

9. What is claimed, and what is not

We separate these deliberately, because the value of the measurements above depends entirely on the precision of the claims they support.

9.1 The boundary that matters most

Everything here concerns the *proof system*. The *transaction* carrying a proof is signed with Ed25519, which is elliptic-curve and falls to Shor. An adversary with a quantum computer does not need to break any relation in this paper: they forge the signature on the envelope and the post-quantum content inside becomes irrelevant. This is a post-quantum component running on a pre-quantum chain, and the missing half is Stellar’s to supply, not ours.

9.2 Claimed

- **Post-quantum in kind:** hash-based, no elliptic curves, no trusted setup. Nothing in the proof system falls to Shor.
- **Post-quantum in strength,** at the deployed configuration: roughly 96 conjectured bits, near 48 against a Grover-assisted adversary. Both numbers rest on the standard conjectured FRI estimate rather than a proven bound.
- **Single use by consensus:** the nullifier is burned in contract storage, so it survives a restart of the server. It does not survive a redeploy of the verifier, which starts a fresh set.
- **Unlinkability across uses:** computational, resting on the hiding of the commitment under a random-oracle-style assumption with a 248-bit blinder.

9.3 Not claimed

- No bespoke security proof for the sponge-capacity split or the compression convention. The compression function is a truncated public permutation without feed-forward, which is *provably* not collision resistant; no forgery follows from that in this system as far as our analysis goes, and both statements are recorded rather than one of them.
- The identifying mode does not hide which credential paid. Only the unlinkable mode does.
- The two halves of the unlinkable proof are composed by a shared public value, not fused into one trace, so on-chain they are two transactions.
- Amounts and recipients are in the clear. This hides *who*, not *how much* or *to whom*.

9.4 On priority

An earlier draft claimed no prior x402 payment gated by an anonymous credential. That was wrong. MicoPay⁷ does the same shape on Stellar, independently and deployed on testnet: an agent buys a credential over x402, spends it with a proof of membership, and a Soroban contract burns the nullifier. We record the error here rather than quietly narrowing the claim elsewhere.

What separates the two is the proof system. MicoPay verifies through the BN254 host functions (`g1_msm`, `pairing_check`); its soundness rests on discrete log over a pairing curve,

⁷github.com/Micopay/micopay-protocol

and it carries a trusted setup. Shor’s algorithm breaks that assumption. The construction here rests on a hash function, carries no setup, and concedes only the factor of two Grover takes out of the exponent.

The narrowed claim is therefore: no prior STARK verified directly by a Soroban contract, and no prior *post-quantum* credential gating an x402 payment. Neither half claims the credential construction is new — the Related work section above places it against the literature it comes from, where it is neither the smallest nor the strongest instance. The claim is about a verifier that runs in consensus rather than in software. Everything else verifying zero-knowledge proofs on Soroban today is pairing-based — MicoPay and StellarVeil over BN254, the SDF privacy-pool prototype over BLS12-381, and RISC Zero reaching Stellar only through Nethermind’s `groth16_verifier.wasm`, a wrapper that reintroduces pairings and a trusted setup at the last step so that no STARK ever touches the chain.

The gap has a structural cause. CAP-0075 (Final, Protocol 25) gives Soroban a native Poseidon2 permutation, but its `field` argument admits only the scalar fields of BLS12-381 and BN254. Circle STARKs are cheap precisely because they live over a 31-bit field, and no host function covers one. A Mersenne-31 verifier must be written out in contract code, beneath the 400M-instruction cap, or it does not exist.

That is a statement about what public search surfaced on 5 August 2026, not a proof of absence. If the remainder is wrong too, the correction belongs in the repository and will be put there.

10. Reproducing

```
git clone https://github.com/Galmanus/pq402 && cd pq402
npm install && (cd cli && npm install)
cp .env.example .env
node setup.mjs # generates and funds both accounts
curl -s https://channels.openzeppelin.com/testnet/gen # a facilitator key

./demo.sh # the identifying loop, end to end
./demo-crowd.sh # the unlinkable one, end to end
```

One step cannot be scripted: the Circle testnet USDC faucet is captcha-gated, and `setup.mjs` prints the address to paste there.

Running `demo-crowd.sh` twice within the same hour-long epoch produces a refusal — from the contract, not from the script. That refusal is the single-use guarantee showing its face rather than being asserted.

Source, transaction hashes and the full measurement log: <https://github.com/Galmanus/pq402>. The Circle-STARK library the credential relation is built from is `riverrun`. MIT licensed.